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Small-scale temporal variability in assemblages of larval fishes: implications for sampling

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Abstract. Short-term changes in the assemblages of larval fishes in waters over the inner continental shelf off Sydney (Australia) were examined. This was done to highlight the potential sampling problem of confounding short-term variability with longer-term and spatial variability in plankton communities. Assemblages at the surface and at 30 m depth were sampled twice a day over consecutive days at two locations 5 km apart. The assemblages and the densities of individual taxa varied greatly between sampling times and days. The small-scale temporal changes in densities of larvae were not the same at both locations or between depths. Large and consistent depth-related differences in assemblages and abundances of most taxa were evident. The consequences of small-scale temporal variations are discussed relative to the design of sampling programs for ecological and environmental purposes. Studies that include the double sampling of some stations are recommended. Greater emphasis needs to be placed on fine-scale oceanography and larval behaviour to determine causative mechanisms of short-term changes in assemblages of larval fishes.

Introduction

Larval fishes and zooplankton are seldom distributed randomly, rather they are most often patchy, being concentrated in time and space (Wiebe, 1970; Haury *et al.*, 1978; Hewitt, 1981; Mackas *et al.*, 1985; Heath, 1992). The temporal patchiness of fish larvae depends on numerous interacting factors, including the time and location of spawning, length of larval life, larval behaviour and physical processes that influence advection, diffusion and/or concentration and transportation (Iles and Sinclair, 1982; Shelton and Hutchings, 1982; Legendre and Demers, 1984; Chenoweth *et al.*, 1989; Leis, 1991; Cowen *et al.*, 1993). Subsequently, larval assemblages can vary considerably over a wide range of temporal scales, ranging from hours to days to seasons and years (Young *et al.*, 1986; Loeb and Rojas, 1987; Moser *et al.*, 1987; Kingsford and Choat, 1989; Davis *et al.*, 1990; Hettler and Chester, 1990; Thorrold *et al.*, 1994a). Thus, descriptions and interpretations of temporal and spatial patterns of the distributions of fish larvae may vary with the time and space scales of the sampling program (see also Omori and Hamner, 1982; Andrew and Mapstone, 1987; Weins, 1989).

Many studies concerning the patterns of distribution of larval fishes in coastal and nearshore waters have been done over large temporal and spatial scales (e.g. Young *et al.*, 1986; Williams *et al.*, 1988; Sabates, 1990; Doyle *et al.*, 1993; Gray, 1993; Moser and Smith, 1993), whilst patterns and processes occurring over smaller scales have often been overlooked (but see Jahn and Lavenberg, 1986; Kingsford and Choat, 1989; Kobayashi, 1989; Thorrold *et al.*, 1994a,b). Some short-term studies have focused on the relationships of larval supply to recruitment of reef-associated fishes (e.g. Milicich *et al.*, 1992; Thorrold *et al.*, 1994a,c) and the effects of oceanographic features on distributions of larvae (e.g. Shanks, 1983; Kingsford *et al.*, 1991). Small-scale temporal variations in planktonic assemblages have the potential to confound many large-scale comparisons and make quantitative

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interpretations of data difficult. For example, in large-scale studies, different locations are usually sampled at different times of the day and/or on different days throughout a sampling cruise (e.g. Young *et al.*, 1986; Olivar, 1990; Sabates, 1990; Gray, 1993). Consequently, any description of spatial pattern, or any inferences of differences in assemblages between locations, is potentially confounded with time (e.g. Morrisey *et al.*, 1992; Thrush *et al.*, 1994). Specifically, any differences between locations may solely be due to the fact that the different locations were sampled at different times. It is generally not logistically feasible to sample all locations at the same time in large-scale studies. Hence a description of how assemblages at each site vary over short time intervals, i.e. within and between consecutive days, is essential.

Ample evidence is available to suffice that small-scale temporal variations are prevalent in assemblages of larval fishes (Conte *et al.*, 1979; Kingsford and Choat, 1989; Steffe, 1989; Thorrold *et al.*, 1994a). Short-term variation may be manifest in the horizontal and vertical planes, and thus it is important to know if such variation is synchronous at different depths in the water column. Repeated sampling of assemblages at different depths in the water column, over short time periods at more than one site, is required to establish generality of patterns. Such information will provide greater insight into processes structuring these communities and may aid interpretation of data collected over longer temporal scales.

In this paper I assess short-term changes in the assemblages of larval fishes in waters over the inner continental shelf off Sydney, Australia. This was done to examine the potential confounding of spatial pattern with small-scale temporal variability. I specifically tested whether short-term changes in assemblages were concurrent at two depths in the water column, as well as at two locations. I discuss my findings relative to the design of ecological and environmental sampling programs on larval fishes and zooplankton.

Method

Study area and sampling procedure

Sampling was done in 70–80 m of water off Port Hacking and Marley. The two locations were ~5 km apart and 3 km off the coast (see Gray, 1993, 1996). Each location was sampled at two different times of the day on two consecutive days; 25 and 26 November 1991. All sampling was done during daylight between 08:00 and 16:00 h (2 h after sunrise and 4 h before sunset, respectively). Samples were collected at the surface (0–2 m) and at 30 m depth (deep) on all occasions using a cylindrical-conical net with an 80 cm diameter mouth, 500 μm mesh in the body and 250 μm mesh in the collecting bag. On each occasion, three replicate 10 min horizontal tows were done at each depth at each site. The deep tows were done using a messenger-operated open-close mechanism (General Oceanics model 1000-DT) to prevent contamination of samples upon deployment and retrieval. A flow meter (General Oceanics model 2030R) was positioned in the mouth of each net to measure the amount of water filtered in each tow. The average volume of water filtered per tow was 250 m^3. It took less than 1 h to complete the sampling at each site. All sampling was done aboard the FRV *Kamala*.

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Samples were preserved in 10% formalin/seawater immediately following capture, and after 2 weeks transferred to 70% alcohol. Larval fishes were sorted from the catches using a binocular microscope and were identified to the lowest taxonomic level possible (usually family). Herein, the term larval fish refers to all small fish caught in the sampling gear, excluding yolk-sac larvae.

Oceanographic conditions during the study period

Throughout the two days of sampling the shelf waters off Sydney were influenced by an East Australian Current eddy. Data collected at the Sydney Water Board's Ocean Reference Station (situated 30 km north of the study area) showed that a southerly flowing current of ~0.2 m s^{-1} persisted on both days. A thermocline was present at both locations on both days and was situated at ~20–25 m depth (hence deep tows were below the thermocline). The temperature of water was 18.5°C at the surface and 15°C at 50 m. A 2 m swell from the northeast and a wind from the northeast of 0.4 m s^{-1} persisted throughout the 2 days of sampling. Generally the strength of the wind increased on each day from 0.4 m s^{-1} in the morning to 1 m s^{-1} by late afternoon.

Analysis of data

Catches of larval fishes were standardized to number caught per 250 m^3 of water filtered. Data were transformed to log $(x + 1)$ prior to analysis.

Differences in the structure of the assemblages of larval fishes between depths, sampling times and locations were determined by non-metric multidimensional scaling and one-way analyses of similarities (Field *et al.*, 1982; Clarke, 1993). Similarity matrices based on the Bray–Curtis similarity measure (Bray and Curtis, 1957) were generated, after which non-metric multidimensional scaling was used to generate two dimensional ordination plots (Clarke, 1993). The ordination plots displayed the inter-relationships among samples, based on the relative abundance of each taxonomic grouping. Essentially, samples in plots that were close together were most similar. The log $(x + 1)$ transform ensured that each taxonomic grouping contributed fairly evenly to each analysis, rather than the few abundant taxa in the samples dominating each analysis (see Clarke and Green, 1988; Clarke, 1993). The extent to which the plots displayed the relationships among samples (i.e. goodness of fit) was determined by a stress coefficient. One-way analyses of similarities (ANOSIM) were used to determine which assemblages differed, and similarity percentage analyses (SIMPER) were used to distinguish the species that were most responsible for determining the depth-related differences between assemblages (see Clarke, 1993). ANOSIM is equivalent to univariate analysis of variance (ANOVA) in that it compares within sample similarity to among sample similarity. The PRIMER program was used for all analyses (see Clarke, 1993).

Analysis of variance was used to examine temporal and spatial differences in the diversity and abundances of larval fishes. The model of the four factor analysis of variance was: Day (Day 1 versus Day 2 – Fixed – as they were consecutive); Location (Port Hacking versus Marley – Random); Time (Time 1 versus Time 2

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and Time 3 versus Time 4 – Random and nested in the factors Day and Location) and Depths (Surface versus Deep – Fixed). Time was nested in factors Day and Location to overcome possible confounding effects of the short travelling time between sites (20 min). Data were tested for homogeneity of variances (Cochran's test) prior to analysis. Heterogeneous data were still analysed because balanced data sets are robust to deviations from homogeneity, but alpha was set at $P = 0.01$ to reduce the risks of Type 1 error (Underwood, 1981).

Results

Variations in assemblages of larval fishes

A total of 2016 individuals (354 at the surface, 1662 at 30 m) of 56 identifiable taxa of larval fishes were caught in the study. A greater total number of taxa and individuals was caught at 30 m than at the surface on all occasions at both sites (Table I). The assemblages of larval fishes at each depth were dominated by similar taxa across all sampling times and at both locations, although their rank orders of abundance varied among sampling times and sites (Table I). This was partly due to the very small numbers of some taxa.

Assemblages of larval fishes varied temporally and spatially. The major delineation between assemblages was that based on depth, i.e. surface versus 30 m, regardless of location and time (Figure 1). All surface samples differed significantly from deep samples (one-way analysis of similarities, $R = 0.700$, $P < 0.001$). The three most characteristic taxa of each depth as determined by the similarity percentages analyses were: surface – *Sardinops neopilchardus*, *Acanthopagrus australis* and *Gerres subfasciatus*, deep – *S.neopilchardus*, Creediidae and *Sillago flindersi*. Creediidae, *Sillago flindersi* and Platycephalidae contributed greatest to the delineation of the surface and deep assemblages (as determined by similarity percentages analysis).

Because of the depth-related differences between assemblages, separate analyses were done on each depth in an attempt to clarify temporal differences among samples. Replicate deep samples for each site/time tended to group together, but this was not as evident for replicate surface samples. Rather, surface samples were highly variable, being widely scattered in the ordination plots (Figure 1), presumably due to the low abundances of larvae in these samples. There was some delineation among surface and deep samples according to day (Day 1 versus Day 2), but there was less obvious separation of samples based on time of day and to site, particularly for surface samples (Figure 1).

Variations in densities of larval fishes

The mean numbers of total taxa, total individuals and the densities of the common taxa varied temporally and spatially (Figures 2–4, Table II). There were, however, several temporally consistent depth-related trends at both locations. Significantly fewer taxa and individuals were caught at the surface than at depth on all occasions, except Time 1 at Port Hacking (Figure 2, Table II). This same trend was also evident for Labridae, *Sardinops neopilchardus*, *Engraulis australis*,

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Table I. The numbers of total taxa, total individuals and the five most abundant taxa of fish larvae caught at the surface and at 30 m at each location (pooled across three replicates) on each sampling occasion over the 2 day sampling period

Day 1 Time 1 Taxa	No.	Time 2 Taxa	No.	Day 2 Time 3 Taxa	No.	Time 4 Taxa	No.
a) Surface							
Port Hacking							
Total taxa	21	Total taxa	12	Total taxa	9	Total taxa	18
Total individuals	148	Total individuals	30	Total individuals	15	Total individuals	31
Sardinops neopilchardus	34	Monacanthidae	4	*Acanthopagrus australis*	5	Pomacentridae	6
Sillago flindersi	14	Platycephalidae	4	*Omobranchus anomalous*	3	Clinidae/Tripterygiidae	4
Monacanthidae	14	*Sardinops neopilchardus*	4	Gobiidae	1	Unidentified type 1	3
Platycephalidae	12	*Pelates quadrilineatus*	3	Unidentified type 1	1	*Omobranchus analogous*	2
Acanthopagrus australis	11	Pomacentridae	3	*Gerres subfasciatus*	1	Monacanthidae	2
Marley							
Total taxa	17	Total taxa	12	Total taxa	11	Total taxa	6
Total individuals	32	Total individuals	48	Total individuals	31	Total individuals	20
Sardinops neopilchardus	7	*Acanthopagrus australis*	16	Clinidae/Tripterygiidae	10	*Acanthopagrus australis*	5
Gobiesocidae	5	*Sardinops neopilchardus*	8	*Pelates quadrilineatus*	6	*Sardinops neopilchardus*	4
Monacanthidae	2	*Girella tricuspidata*	5	*Acanthopagrus australis*	3	Clinidae/Tripterygiidae	4
Monodactylus argenteus	2	Monacanthidae	5	*Omobranchus anomalous*	3	Gobiesocidae	3
Acanthopagrus australis	1	*Sillago flindersi*	4	Monacanthidae	2	*Sillago flindersi*	2
b) 30 m							
Port Hacking							
Total taxa	30	Total taxa	23	Total taxa	28	Total taxa	28
Total individuals	225	Total individuals	309	Total individuals	216	Total individuals	145
Sillago flindersi	49	*Sillago flindersi*	78	*Sillago flindersi*	32	*Sardinops neopilchardus*	23
Creediidae	21	*Sardinops neopilchardus*	65	Labridae	27	*Hyperlophus vittatus*	20
Triglidae	20	Platycephalidae	44	*Hyperlophus vittatus*	24	Gobiesocidae	14
Centropogon australis	18	Creediidae	26	Creediidae	23	*Sillago flindersi*	12
Sardinops neopilchardus	16	*Neosebastes* spp.	22	*Sardinops neopilchardus*	21	Pomacentridae	12
Marley							
Total taxa	25	Total taxa	28	Total taxa	20	Total taxa	23
Total individuals	124	Total individuals	358	Total individuals	118	Total individuals	166
Creediidae	31	Creediidae	75	*Sardinops neopilchardus*	22	Creediidae	32
Sardinops neopilchardus	31	Triglidae	45	Gobiesocidae	17	*Sardinops neopilchardus*	28
Triglidae	13	*Sardinops neopilchardus*	39	Creediidae	14	Labridae	15
Pomacentridae	8	*Sillago flindersi*	27	*Neosebastes* spp.	11	Pomacentridae	12
Sillago flindersi	7	*Neosebastes* spp.	23	Gobiidae	10	*Sillago flindersi*	12

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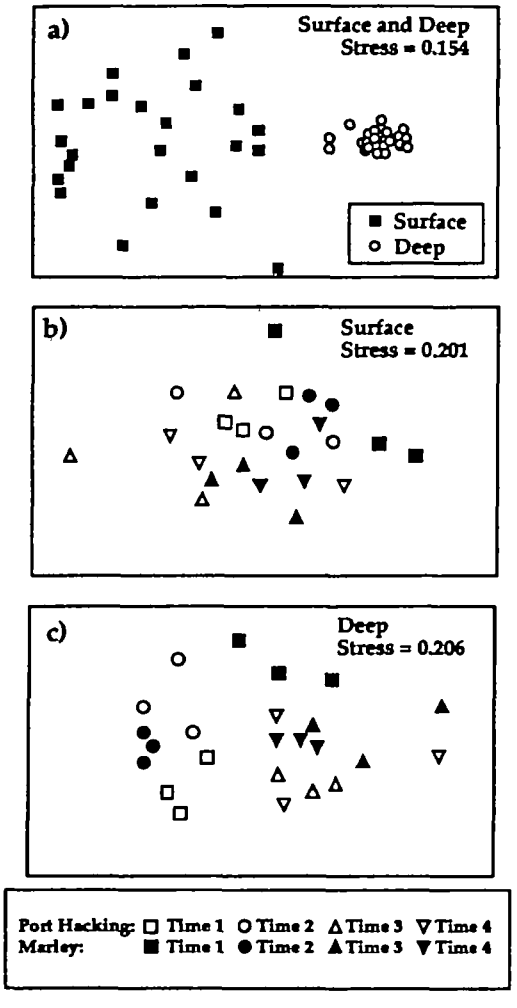

Fig. 1. Non-metric multidimensional scaling ordination plots displaying inter-relationships among assemblages of larval fishes (samples) based on log $(x + 1)$ transformed abundance data of taxonomic groupings. (**a**) Differences between assemblages of larval fishes in surface and deep (30 m) depth strata. (**b**) Differences among surface assemblages collected in each of the four time periods. (**c**) Differences among deep assemblages collected in each of the four time periods.

Centropogon australis, and Platycephalidae (Figures 2–4, Table II). Densities of *Sillago flindersi*, Creediidae, Triglidae, *Hyperlophus vittatus*, Percophididae, *Neosebastes* spp. and Callionymidae were greatest at depth across all times and at both locations (Figures 2–4, Table II). Conversely, no consistent depth-related differences in densities were evident for Monacanthidae, Gobiesocidae, Gobiidae, Pomacentridae and *Acanthopagrus australis* (Figure 4, Table II).

It was evident at both locations, and at both depths, that the numbers of taxa and individuals caught and the densities of most of the common taxa varied

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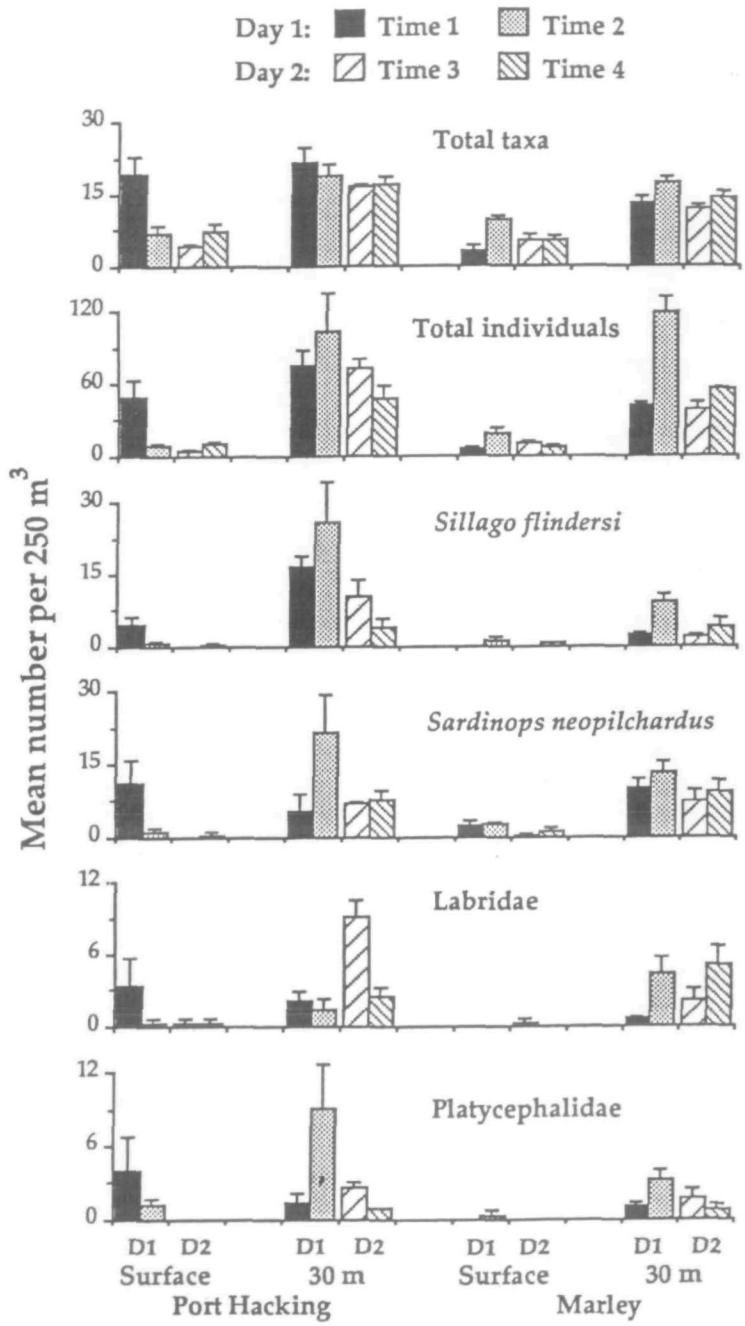

Fig. 2. Mean (+ 1 SE) numbers of larval fishes caught at the surface and at 30 m at each location throughout the 2 days of sampling. D1 = Day 1, D2 = Day 2.

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between days and between times within each day. Moreover, these temporal changes in densities were not always of the same magnitude or direction at both locations or at both depths at a location. For example, densities of Creediidae at Marley doubled between Time 1 and 2, whereas at Port Hacking they did not change significantly over the same period. In contrast, several taxa, including the Gobiidae, Pomacentridae, Callionymidae and Monacanthidae displayed no significant differences in densities between times, albeit they occurred in low densities (Figure 4, Table II).

There were no consistent differences between locations at either depth in either the numbers of taxa and individuals caught nor in the densities of the common taxa over the 2 day period (Figures 2–4, Table II). In most cases, the number of larvae caught at a location was dependent on both the time of day (although the rank of time of day, i.e. early versus late, was not consistent) and the day sampled. For the majority of the common taxa, variations in densities at each location between sampling times were equivalent to differences between locations at any given time.

Discussion

The assemblages and densities of larval fishes at each location and depth varied throughout the 2 day sampling period. Moreover, these variations were not synchronized at different depths in the water column or at two relatively close locations. Such short-term variations were not unexpected: similar trends have been reported for larval fishes elsewhere (Conte *et al.*, 1979; Kingsford and Choat, 1989; Steffe, 1989; Lyczkowski-Shultz *et al.*, 1990; Thorrold *et al.*, 1994a). For example, Kingsford and Choat (1989) documented how the densities of small fishes in surface waters fluctuated throughout a 3 day period and that such variations were not concurrent at different, but nearby localities. An understanding of this short-term variation assisted in the interpretation of the broad-scale patterns that took 3–4 days to sample. These combined results highlight how any description of an assemblage and any estimate of the density of individual taxa at a particular location may depend on the time of sampling. This may confound any comparisons of assemblages and/or densities of larvae between locations, and also over longer temporal scales. The observed fluctuations in the densities of most larvae in this short-term study were generally much less than those observed over longer (seasonal) temporal periods (Gray, 1995). Nonetheless, short-term changes have important implications for many types of field studies concerning the dynamics of planktonic organisms, particularly those that employ a sampling program that samples different locations at different times.

The results presented here illustrate how caution must be exercised when interpreting large-scale comparisons where sampling occurs over a number of days. Clearly, the outcomes of any comparisons of assemblages and/or densities of larvae between locations may depend on when the samples were collected [and on precisely where they were collected, i.e. small-scale spatial patchiness (Houde and Lovdal, 1985; Davis *et al.*, 1990; Thorrold *et al.*, 1994b; Gray, 1995)]. For example, if both locations in the present study had been sampled at Time 1 it

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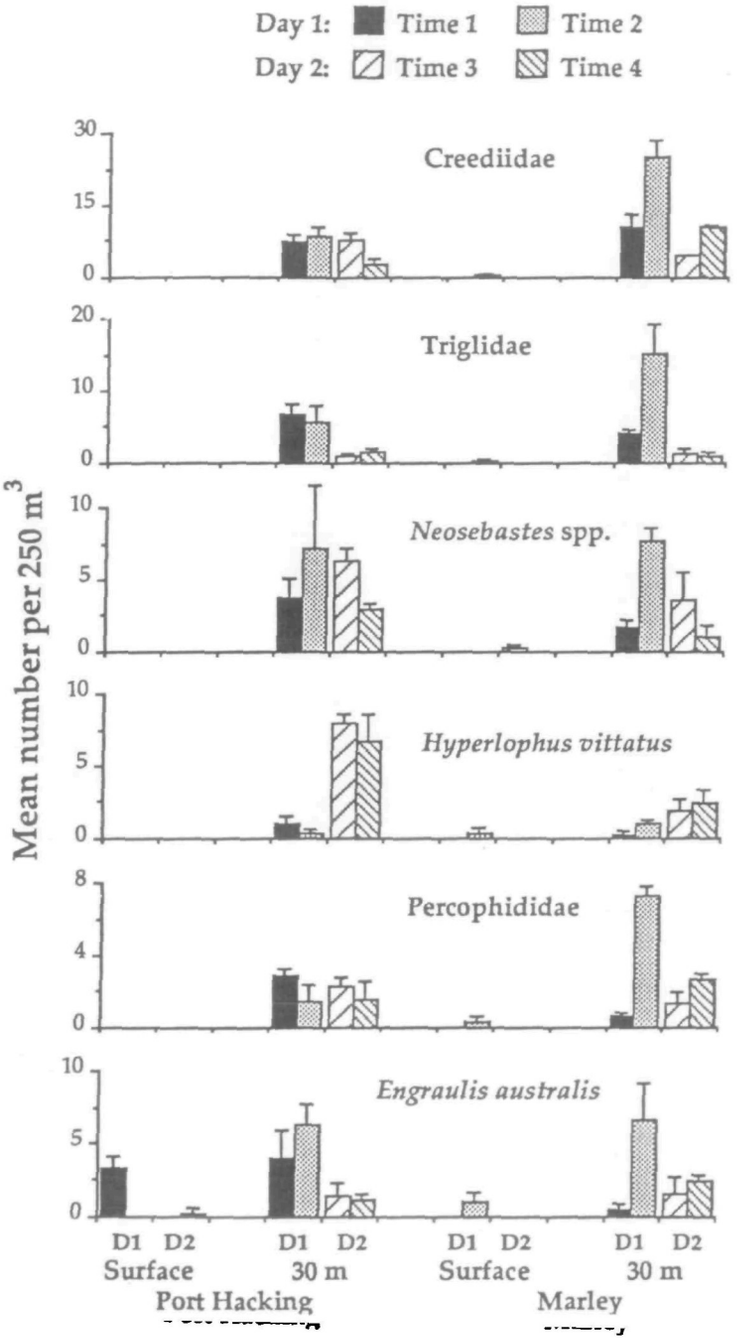

Fig. 3. Mean (+ 1 SE) numbers of larval fishes caught at the surface and at 30 m at each location throughout the 2 days of sampling. D1 = Day 1, D2 = Day 2.

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Table II. Summary of analyses of variance of the numbers of larval fishes caught at the surface and at 30 m at Port Hacking and Marley throughout the two days of sampling

	Source of variation										
	Da	L	Da × L	T(Da + L)	De	Da × De	L × De	Da × L × De	T(Da + L) × De	Residual	Cochran's test
	Degrees of freedom in F-test										
	(1, 1)	(1, 4)	(1, 4)	(4, 32) (4, 36)[a]	(1, 1)	(1, 1) (1, 37)[b]	(1, 4) (1, 36)[a]	(1, 4) (1, 36)[a]	(4, 32)	(32) (36) (37)[c]	
Total taxa	0.100 ns	0.430 ns	0.228 ns	0.627***	11.118 ns	0.030 ns	0.084 ns	0.501 ns	0.232***	0.054	ns
Total individuals	2.690 ns	0.608 ns	0.547 ns	1.167***	34.205*	0.102 ns	0.022 ns	0.697 ns	0.826***	0.108	ns
S.flindersi	5.329 ns	4.918 ns	1.649 ns	0.982**	28.160 ns	0.191 ns	1.251 ns	0.035 ns	0.729*	0.200	ns
S.neopilchardus	5.088 ns	0.017 ns	0.204 ns	0.191 ns	24.557*	1.549 ns	0.108 ns	0.574 ns	1.846***	0.294	ns
Labridae	0.505 ns	0.903 ns	0.010 ns	0.871*	11.916 ns	1.679 ns	0.164 ns	0.752 ns	0.745*	0.250	ns
Platycephalidae	3.346 ns	1.952 ns	0.940 ns	0.404 ns	6.677 ns	0.066 ns	0.063 ns	0.460 ns	0.853**	0.156	ns
E.australis	3.200 ns	0.565 ns	0.942 ns	1.037**	8.789*	0.042 ns	0.026 ns	0.187 ns	0.860**	0.190	ns
Creediidae	1.284 ns	1.095 ns	0.104 ns	0.485***	54.799 ns	0.864 ns	0.711 ns	0.014 ns	0.329**	0.073	ns
Triglidae	5.610 ns	0.041 ns	0.213 ns	0.320*(*)	22.139**	4.787 ns	0.001 ns	0.079 ns (nse)	0.152 nse	0.108 (0.113)	ns
H.vittatus	3.455 ns	0.520 ns	1.024 ns	0.145 ns (ns)	11.358 ns	4.316 ns	0.884* (**)	0.629* (*)	0.049 nse	0.094 (0.089)	ns
Neosebastes spp.	0.078 ns	0.229 ns	0.304 ns	0.507 ns	23.699 ns	0.208 ns	0.427 ns	0.529 ns	0.395 ns	0.231	*
Percophididae	0.158 ns	0.116 ns	0.043 ns	0.850***	12.899*	0.047 ns	0.026 ns	0.001 ns	0.585***	0.118	ns
C.australis	2.768 ns	0.956 ns	0.278 ns	1.407***	9.043*	0.219 ns	0.048 ns	0.447 ns	0.482**	0.108	ns
Pomacentridae	0.000 ns	0.781 ns	1.001 ns	0.874 ns (ns)	7.505 ns	0.080 ns	0.985 ns (ns)	0.026 ns (nse)	0.165 nse	0.400 (0.374)	ns
Gobiesocidae	0.869 ns	0.163 ns	0.267 ns	0.050 ns (ns)	3.517 ns	3.662 ns	0.700 ns (ns)	0.331 ns (ns)	0.133 nse	0.253 (0.240)	ns
Monacanthidae	0.420 ns	0.227 ns	0.207 ns	0.341 ns (ns)	1.000 ns	1.842 ns (**)	0.607 ns (ns)	0.390 ns (nse)	0.143 nse	0.237 (0.226)	ns
Gobiidae	0.134 ns	0.030 ns	0.039 ns	0.460 ns	6.474 ns	0.717 ns	1.155 ns	0.015 ns	0.461 ns	0.208	ns
A.australis	0.538 ns	0.048 ns	0.224 ns	0.436 ns	6.294 ns	0.228 ns	0.153 ns	0.047 ns	0.885**	0.164	ns
Callionymidae	0.121 ns	0.028 ns	0.033 ns	0.692*	9.453**	0.287 ns	0.000 ns	0.137 ns	0.520*	0.176	ns

Da, Day; L, Location; T, Time: De, Depth; df, degrees of freedom: *,**,***, significant at $P < 0.05$, $P < 0.01$, $P < 0.001$, respectively; ns, not significant at $P > 0.05$; nse, not significant at $P > 0.25$ and term eliminated and pooled.

[a] When the T(Da × L) × De interaction term was not significant at $P > 0.25$ it was eliminated to give more powerful tests of the Da × L × De interaction term (1, 36 df) and the L × De interaction term (1, 36 df) and the T(Da × L) term (4, 36 df).

[b] When the Da × L × De interaction term was not significant at $P > 0.25$ following elimination of the T(Da × L) × De term it was eliminated to provide a more powerful test of the Da × De interaction term (1, 37 df).

[c] New Mean Square with either 36 df (a) or 37 df (b).

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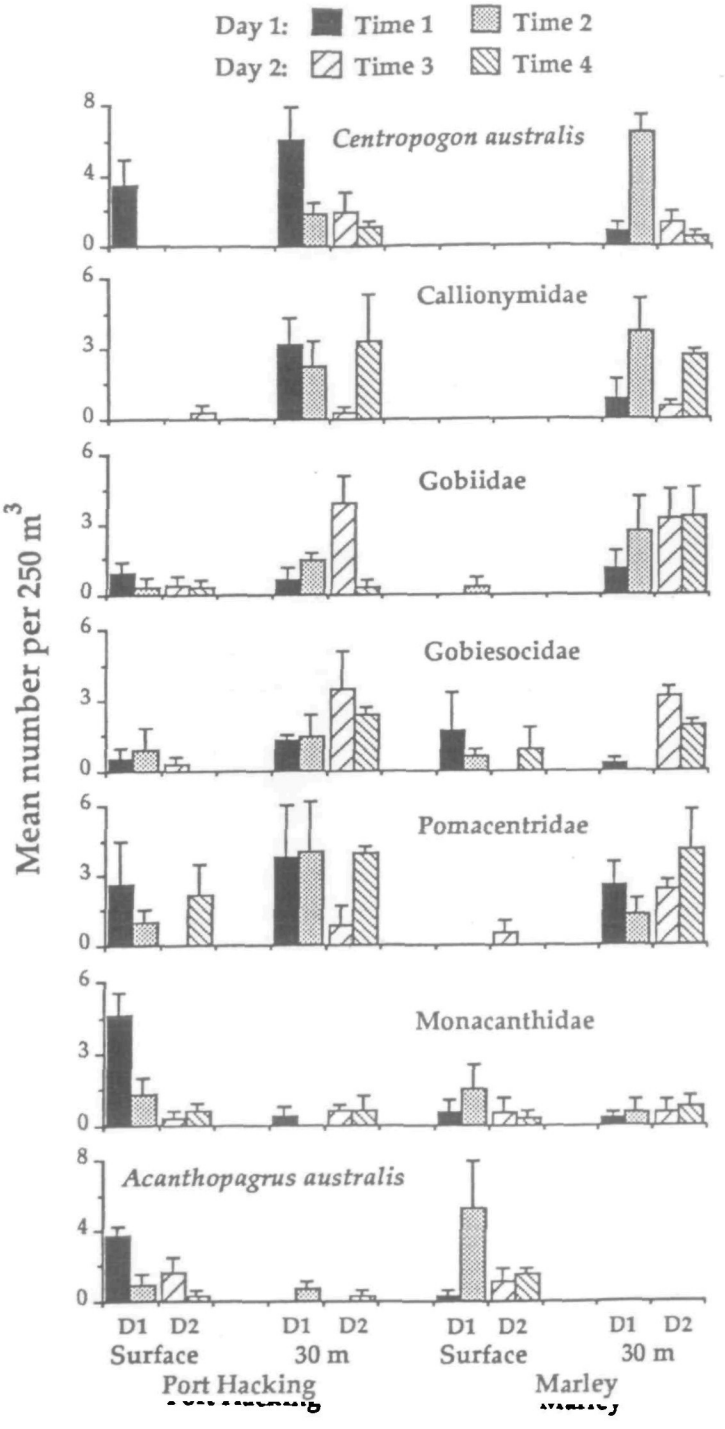

Fig. 4. Mean (+ 1 SE) numbers of larval fishes caught at the surface and at 30 m at each location throughout the 2 days of sampling. D1 = Day 1, D2 = Day 2.

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would be concluded that more taxa and individual fish larvae occurred at both depths at Port Hacking than at Marley. If, however, Port Hacking had been sampled at Time 4, but Marley at Time 1, then it would have been concluded that there were no differences in the numbers of individuals between the two locations at either depth (see Figure 2). Similar scenarios were evident for most taxa and this also applied to their vertical distributions; for example, many taxa occurred in greatest densities at depth except at Time 1 Port Hacking. The data also suggested that even if samples at different locations and depths were sampled simultaneously, any differences between them may not necessarily be consistent through time. That is, the temporal trajectories of assemblages at different locations and depths may not necessarily be the same. This was true over longer time periods in these waters (Gray, 1993, 1995), and may particularly be the case where dynamic and variable oceanographic conditions prevail. These results demonstrate the need for temporal and spatial replication in all types of sampling programmes and further highlights the difficulties in detecting anthropogenic impacts, such as effluent plumes from sewage outfalls, on these assemblages (see also Gray *et al.*, 1992; Gray, 1993). The results also illustrate the need to have multiple control locations in any environmental impact study (see Underwood, 1991, 1992, 1994). Sampling schedules that incorporate double sampling of some locations may need to be included in some broad-scale studies. Repeated sampling among times may be the only way to reveal large spatial patterns, but the duration of time that larvae of different species are present should also be considered.

Despite the variation depicted above, there were no major changes in the taxonomic composition of the ichthyofauna at either location over the two sampling days. This indicated that no new assemblage was transported into the study area over this time, which suggests that the patch size of this assemblage was relatively large and the spawning output of adults was temporally concordant, or that the fish larvae were being retained in the study area (e.g. Smith and Morse, 1985). Fine-scale circulation patterns of the waters in the study area were not available during this period, but data collected at the Ocean Reference Station suggested that 24 km of water may have passed through the study area over the 2 days of sampling (33 h). Larval fishes may be able to maintain their positions in shelf waters by behavioural movements (e.g. vertical migration), taking advantage of different currents (e.g. Norcross and Shaw, 1984; Cowen *et al.*, 1993). Postflexion fishes would probably be better able to maintain their positions than preflexion and flexion staged fishes as their sensory and swimming capabilities would be relatively better developed (e.g. Blaxter, 1986). However, most fish larvae caught in this study were preflexion. The rate of change in assemblages in a given area would obviously depend on the following: the size of the larval patch, the spawning effort of adults, larval behaviour and oceanographic processes which influence movement and dispersion.

The relatively consistent differences between depths in the assemblages and densities of larval fishes found here concur with the daytime vertical stratification of assemblages in these waters as reported earlier (Gray *et al.*, 1992; Gray, 1993, 1996). The greater densities of many fish larvae at depth and the sparsity of fish

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larvae in surface waters are also in general agreement with those reported for these waters at other times (Gray *et al.*, 1992; Gray, 1993, 1996), but large numbers of fish larvae (e.g. Cheilodactylidae, Mullidae, Scorpididae) do periodically occur in surface waters (Gray, 1993, 1996). The increased densities of many fish larvae at the surface at Port Hacking at Time 1 may have been the result of localized mixing of surface and sub-surface waters due to a breakdown or change in the thermocline (e.g. due to an internal wave). Although this was not detected in the present sampling, short-term and small spatial scale changes in the position of the thermocline are common in the shelf waters off Sydney (Lee and Pritchard, 1993; Gray, 1995, 1996). It was unlikely that this sample contained the remnants of larvae that had migrated to surface waters during the night as it was collected 5 h after sunrise. Moreover, this anomaly was not observed at other sampling times which occurred at earlier times of the day (i.e. Time 1 Marley and Time 3 Port Hacking). Similar episodic increases in the abundances of typically deep-dwelling fish larvae in surface waters have been observed at other times and places in the continental shelf waters off Sydney (Gray, 1995).

In conclusion, it is obvious that small-scale temporal fluctuations play an important part in the dynamics of assemblages of larval fishes in continental shelf waters. Such fluctuations need to be addressed and incorporated into future sampling designs for broad-scale studies on fish larvae. This may be necessary to avoid confounding small-scale temporal differences with broad-scale spatial differences. Small-scale variations do, however, pose interesting studies in themselves. The challenge lies ahead to determine what processes drive these small-scale variations. This will entail a more detailed knowledge of the behaviour of larvae, the distributions of prey and predators and fine-scale oceanography.

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